



DevOps Project by Dayyan Hasan

Deploying a Web Application on AWS

From bare-metal VPC networking to load-balanced auto-scaling compute.

ARCHITECTURE • NETWORKING • HIGH AVAILABILITY

The Multi-Tenant Cloud Problem

⚠ The Danger: Public Sidewalk

Public clouds share servers among millions. Launching databases directly without an isolated network is like placing company secrets on an open sidewalk.

0.0.0.0/0 Exposed to whole web

🛡 The Solution: VPC Isolation

A Virtual Private Cloud (VPC) carves out your private digital fence inside AWS. By default, nobody gets in and nobody gets out.

10.0.0.0/16 Private Gated Space

The Gated Community Analogy



VPC

The Outer Perimeter

You buy an empty piece of land and erect a perimeter security wall. Nothing enters without explicit permission.



Public Subnet

The Visitor Gatehouse

Positioned at the entrance. Welcomes external road traffic. Houses load balancers and front-door proxies.



Private Subnet

The Inner Homes

Located deep inside. No outside road access. Ideal for application instances and persistent data stores.

Public vs. Private Subnets

Feature	Public Subnet	Private Subnet
Direct Route to IGW	Yes (0.0.0.0/0 → igw)	No (Isolated)
Public IP Assigned	Yes (Auto-assign Enabled)	No (Private IP Only)
Primary Purpose	Load Balancer, NAT, Bastion	Backend App Servers, Databases
Outbound Internet Path	Direct via Internet Gateway	Through NAT Gateway only

Gateways: Two-Way vs One-Way

IGW Internet Gateway

Two-Way Door. Translates public addresses and routes traffic in and out for public subnets.

Inbound: Allows users to view the site

Outbound: Allows public resources out

Target: `igw-0825b5dbfb5692b8a`

NAT NAT Gateway

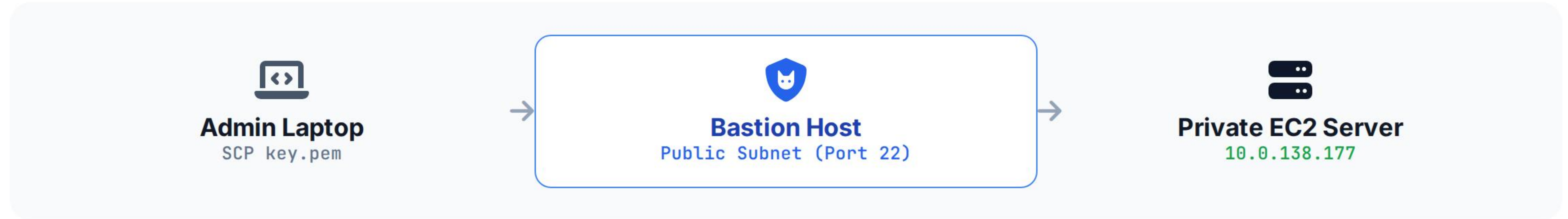
One-Way Mask. Deployed in public subnets so private EC2s can download OS updates safely.

Inbound: Completely blocks unknown requests

Outbound: Permits internal servers to reach apt/yum

Stateful masquerading with Elastic IP

The Bastion Host Pattern



Step 1: Jump Box Entry

SSH only into the single hardened Bastion in the public subnet using your private key.

Step 2: Internal Hop

From the Bastion, initiate an internal hop to private instances via their internal IP addresses.

High Availability Across Two AZs

An **Availability Zone (AZ)** is one or more distinct data centers isolated by power, cooling, and water supplies.

Golden Rule of Production

Always mirror your subnets across at least two Availability Zones (**ap-south-1a** and **ap-south-1b**).



Balancing & Dynamic Scaling



Application Load Balancer

The **Traffic Cop**. Sits at the public edge listening on Port 80 and evenly fans out requests to healthy instances.

Port 80 Ingress → Port 8000 Forward
Continuous Target Group Health Checks



Auto Scaling Group (ASG)

The **Workforce Manager**. Launches identical EC2 instances when traffic surges and terminates them when idle.

Desired Capacity: 2 Instances
Max Scaling Elasticity: Up to 4

The User Journey (Port 80 to 8000)

STEP 1

User Request

Browser calls public ALB DNS hostname on Port 80.



STEP 2

Internet Gateway

IGW receives packet and directs it to public subnets.



STEP 3

ALB Evaluates

Checks Target Group health checks and selects instance.



STEP 4

Private Delivery

Packet delivered to private EC2 serving Port 8000.

Security Groups: Least Privilege

ALB Security Group

Only listens to external clients.

Inbound: TCP 80 from 0.0.0.0/0

Outbound: All traffic allowed

Backend EC2 Security Group

Explicitly references trusted security groups instead of IP ranges.

Inbound: TCP 8000 (Source: ALB-SG only)

Inbound: TCP 22 (Source: Bastion-SG only)

Architecture Cheat Sheet

Screenshot this for study prep!

VPC	Private cloud network sandbox	Multi-AZ	Physical data center fault isolation
IGW	2-way front door for public subnets	NAT Gateway	1-way outbound updates for private tier
Bastion Host	Secure SSH jump box for backend access	ALB	Distributes HTTP/S across instances
Target Group	Pool of healthy backend targets	ASG	Dynamic autoscaling compute fleet



Thanks for learning with me!

Follow Along for More DevOps

Connect with Dayyan Hasan for hands-on cloud architectures,
system design guides, and step-by-step tutorials.

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